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(57) Abstract:

ABSTRACT LOCATION-FILTERED SERVICE PLATFORM WITH BIDIRECTIONAL RATING AND IMAGE VERIFICATION The present disclosure provides a system for connecting daily wage workers with job opportunities. The system includes a server (100) configured to execute instructions stored in a memory. The server includes a user registration module (102, 104, 106, 108) configured to receive registration data from users classified as either workers or clients, wherein worker registration data includes an identity document image. The server includes a database (114) configured to store identity document images associated with each worker. The server includes a job assignment module (306) configured to enable assignment of workers to jobs. The server includes an identity verification module (312, 314, 316, 318) configured to receive a captured image of a worker at a job location from a client device, cross-verify it with the stored identity document image, and generate a verification confirmation upon successful cross-verification. (FIG. 1)

FORM-2

**THE PATENTS ACT, 1970
(39 of 1970)**

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**&
THE PATENTS RULES, 2003**

**COMPLETE
SPECIFICATION
(See section 10 and rule 13)**

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TITLE OF INVENTION

**LOCATION-FILTERED SERVICE PLATFORM WITH
BIDIRECTIONAL RATING AND IMAGE VERIFICATION**

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**THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES
THE INVENTION AND THE MANNER IN WHICH IT IS TO BE
PERFORMED**

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CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to digital employment platforms for connecting workers with job opportunities, and more particularly to a system and
10 method for connecting daily wage workers with clients using dual-layer identity verification comprising Aadhaar-based document validation and on-site facial recognition cross-verification.

BACKGROUND

[0004] The following text of background art is provided for purposes of
15 understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] The informal labor market constitutes a substantial portion of the global workforce, with millions of daily wage workers depending on temporary employment for their livelihood. Digital employment platforms have emerged to
20 address inefficiencies in traditional hiring methods. Platforms such as LinkedIn, Indeed, and Upwork have streamlined job searches for skilled professionals by providing structured interfaces for job posting, application management, and candidate discovery. Government initiatives including India's e-Shram portal and the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)
25 have attempted to provide structured employment opportunities for unskilled and semi-skilled laborers. Mobile applications such as WorkIndia and QuikrJobs have been developed to serve blue-collar workers by offering job listings and application functionalities. These existing platforms and initiatives represent the current state of digital employment solutions in the labor market.

30 [0006] Despite these technological advancements, several technical problems persist in connecting daily wage workers with employment opportunities. First,

existing platforms predominantly cater to white-collar professionals and fail to address the distinct requirements of unskilled and semi-skilled laborers, including digital literacy barriers, limited smartphone access, and absence of localized job listings. Second, government initiatives such as e-Shram and MGNREGA face inefficiencies including job allocation delays, lack of transparency, and bureaucratic obstacles, while lacking features such as real-time job notifications and instant hiring mechanisms. Third, mobile employment applications for blue-collar workers lack real-time job matching capabilities, verified employer-worker interactions, and secure payment mechanisms. Fourth, a trust deficiency exists in the informal employment sector due to the absence of comprehensive identity verification systems, leading to concerns regarding worker authenticity, impersonation, and fraud. Fifth, existing platforms do not incorporate dual-layer verification mechanisms that combine document-based identity validation with on-site biometric cross-verification to establish trust between workers and employers. Sixth, inefficient filtering options based on location, skills, and wage expectations hinder job discovery for workers seeking employment within their geographic proximity.

[0007] It has been appreciated that a system and method is needed that overcomes one or more of these problems.

20 [0008] The present invention addresses the above shortcomings of the prior art. However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

SUMMARY

[0009] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0010] In a first aspect, a system for connecting daily wage workers with job opportunities is provided. The system comprises a server configured to execute instructions stored in a memory, the server comprising: a user registration module configured to receive registration data from a plurality of users, wherein each user
5 is classified as either a worker or a client, wherein for workers the registration data comprises personal details, work category information, a profile picture, and an Aadhaar card image for identity validation, and wherein for clients the registration data comprises client details and profile information; an authentication module configured to perform OTP-based phone number verification for each user, encrypt
10 user credentials, store encrypted credentials in a database, assign role-based access control based on user classification, and generate a JSON Web Token for authenticated sessions; a job management module configured to receive job posting data from clients, wherein the job posting data comprises job title, description, wage, location, job type, gender preference, and working hours, assign a unique job
15 identifier to each job posting, store job postings in the database, and trigger notifications to workers matching job criteria; a job discovery module configured to display job listings to workers, apply a location-based filter to display jobs within a worker's city by default, receive filter parameters from workers comprising wage range, job type, and region, display filtered job listings based on received filter
20 parameters, and receive job applications from workers; a job assignment module configured to enable clients to review worker profiles and shortlist candidates, send selection notifications to selected workers, and receive acceptance responses from workers; an identity verification module configured to receive a captured photo of a worker at a job location from a client device, cross-verify the captured photo with
25 the Aadhaar card image stored during worker registration, generate a verification confirmation upon successful cross-verification, flag the worker profile upon verification failure, and update job status upon successful identity verification; a job completion module configured to receive task completion proof from workers, receive a verification code from clients, validate the verification code, mark the job
30 as completed upon successful validation, and flag the job as pending confirmation upon validation failure; a payment processing module configured to record cash

payments or process digital payments via an in-app payment gateway, update worker earnings, and send payment notifications to workers; and a review module configured to receive ratings and reviews from clients regarding worker performance and receive ratings and reviews from workers regarding client behavior, wherein the ratings and reviews are stored in the database and associated with respective user profiles.

[0011] The system provides a comprehensive digital platform that addresses multiple technical problems in the informal labor market. The dual-layer identity verification mechanism combining Aadhaar-based document validation with on-site photo cross-verification establishes trust between workers and clients by reducing impersonation and fraud. The location-based filtering system with default city-level job display reduces information overload for workers while maintaining flexibility for broader searches. The role-based access control and JWT-based authentication ensure secure user sessions and prevent unauthorized access to platform functionalities. The bidirectional review system promotes accountability and transparency by enabling both workers and clients to evaluate each other, thereby building credibility within the platform ecosystem.

[0012] In a second aspect, a method for connecting daily wage workers with job opportunities using dual-layer identity verification is provided. The method comprises: receiving, by a server, registration data from a user, wherein the registration data comprises personal details, work category information, a profile picture, and an Aadhaar card image when the user is classified as a worker; performing OTP-based phone number verification for the user; encrypting user credentials and storing encrypted credentials in a database; assigning role-based access control based on user classification and generating a JSON Web Token for authenticated sessions; receiving, from a client, job posting data comprising job title, description, wage, location, job type, and working hours; assigning a unique job identifier to the job posting and storing the job posting in the database; displaying job listings to workers with a location-based filter applied to display jobs within a worker's city by default; receiving filter parameters from a worker and displaying filtered job listings based on received filter parameters; receiving a job

application from the worker; enabling the client to review worker profiles and select a worker for the job; sending a selection notification to the selected worker; receiving, from a client device, a captured photo of the worker at a job location; cross-verifying the captured photo with the Aadhaar card image stored during
5 worker registration; generating a verification confirmation upon successful cross-verification and updating job status; receiving task completion proof from the worker; receiving a verification code from the client and validating the verification code; marking the job as completed upon successful validation; processing payment via cash recording or digital payment gateway; updating worker earnings and
10 sending payment notifications; and receiving ratings and reviews from the client regarding worker performance and from the worker regarding client behavior.

[0013] The method provides a structured workflow for connecting daily wage workers with employment opportunities while addressing trust deficiencies in the informal labor market. The dual-layer identity verification comprising Aadhaar-
15 based registration and on-site photo cross-verification ensures worker authenticity and reduces the risk of impersonation. The location-based filtering mechanism streamlines job discovery by presenting workers with geographically relevant opportunities. The verification code mechanism for job completion confirmation ensures accountability and prevents disputes regarding task completion status.

[0014] In a third aspect, a non-transitory computer-readable medium storing instructions is provided. The instructions, when executed by a processor, cause the processor to perform operations comprising: receiving registration data from a user, wherein the registration data comprises personal details, work category information, a profile picture, and an Aadhaar card image when the user is classified
25 as a worker; performing OTP-based phone number verification for the user; encrypting user credentials and storing encrypted credentials in a database; assigning role-based access control based on user classification and generating a JSON Web Token for authenticated sessions; receiving job posting data from a client and storing the job posting in the database; displaying job listings to workers
30 with a location-based filter applied by default; receiving a job application from a worker; enabling a client to select a worker for a job and sending a selection

notification; receiving a captured photo of the worker at a job location from a client device; cross-verifying the captured photo with the Aadhaar card image stored during worker registration; generating a verification confirmation upon successful cross-verification; receiving task completion proof and a verification code; marking
5 the job as completed upon successful verification code validation; processing payment and updating worker earnings; and receiving ratings and reviews from the client and the worker.

[0015] The non-transitory computer-readable medium provides a tangible implementation of the dual-layer identity verification system that can be deployed
10 across various computing platforms. The stored instructions enable consistent execution of the verification workflow, ensuring reliable identity confirmation and job completion tracking across different hardware configurations. The combination of Aadhaar-based registration verification and on-site photo cross-verification provides a robust mechanism for establishing trust in worker-client interactions
15 within the informal labor market.

[0016] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

20 **[0017]** The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

25 **[0018]** FIG. 1 illustrates a flowchart for a user registration and authentication method, according to aspects of the present disclosure.

[0019] FIG. 2 illustrates a flowchart for a job posting and discovery method, according to an embodiment.

[0020] FIG. 3 illustrates a flowchart for a job assignment and identity
30 verification method, according to aspects of the present disclosure.

[0021] FIG. 4 illustrates a flowchart for a job completion and review method, according to an embodiment.

DETAILED DESCRIPTION

[0022] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

10 [0023] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements. Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

15 [0024] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

20 [0025] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0026] FIG. 1 illustrates a flowchart for a user registration and authentication method 100 according to various embodiments. The method 100 provides a dual-path registration flow that accommodates different user types while maintaining
25 consistent security measures across both registration paths.

[0027] The method 100 begins with a step 102, where a registration process is initiated. The method 100 then proceeds to a step 104, where a determination is made as to whether a user is registering as a worker or as a client. Based on this determination, the method 100 branches into one of two distinct registration paths.

30 [0028] When the user selects worker registration at step 104, the method 100 proceeds along a first path. A step 106 involves entering personal details and work

category information. During step 106, the worker provides personal details including name, phone number, gender, location, and work category. The worker specifies a skill set to improve job matching accuracy. The worker sets an availability status to indicate when the worker is actively looking for jobs.

5 **[0029]** The method 100 then moves to a step 108, where the worker uploads a profile picture and an Aadhaar card for identity verification. The Aadhaar card upload forms part of a two-factor verification system that ensures transparency regarding personal details of the worker, such as date of birth and address. The Aadhaar card provides identity validation that establishes trust between workers
10 and clients within the platform.

[0030] Following step 108, a step 110 involves sending a one-time password (OTP) to a phone number provided by the worker. The OTP-based verification enhances security and prevents unauthorized access during user registration. The method 100 then reaches a step 112, where a determination is made as to whether
15 the OTP has been verified. If the OTP is not verified, the method 100 returns to step 110 to resend the OTP. If the OTP is verified, the method 100 proceeds to a step 114, where credentials are encrypted and stored in a database. During step 114, passwords and sensitive user data are securely stored using encryption techniques. The encryption techniques include bcrypt password hashing, which provides secure
20 storage of user credentials.

[0031] The method 100 then moves to a step 116, where role-based access control is assigned and a JSON Web Token (JWT) is generated. The JWT provides stateless user sessions and forms part of a secure authentication mechanism. The role-based authorization ensures that workers have access to worker-specific
25 functionality within the platform. The worker registration path concludes at a step 118, where the registration is complete.

[0032] When the user selects client registration at step 104, the method 100 proceeds along a second path. A step 120 involves entering client details and creating a profile. The client provides details including name, contact information,
30 and location, and creates a profile for business or personal use.

- [0033]** The method 100 then moves to a step 122, where an OTP is sent to a phone number provided by the client. The OTP-based verification for clients provides the same security enhancement as for workers, preventing unauthorized access during registration. The method 100 reaches a step 124, where a determination is made as to whether the OTP has been verified. If the OTP is not verified, the method 100 returns to step 122 to resend the OTP. If the OTP is verified, the method 100 proceeds to a step 126, where credentials are encrypted and stored in the database. During step 126, passwords and sensitive user data are securely stored using bcrypt encryption techniques.
- [0034]** The method 100 then moves to a step 128, where role-based access control is assigned and a JWT is generated. The role-based authorization ensures that clients have access to client-specific functionality, such as job posting capabilities. The client registration path concludes at a step 130, where the registration is complete.
- [0035]** The dual-path registration flow of the method 100 differentiates between worker and client registration paths while maintaining consistent security measures. Both paths incorporate OTP-based phone verification, credential encryption using bcrypt, and token-based authentication using JWT. The worker registration path includes additional identity verification through Aadhaar card upload, which provides a two-factor verification system that builds trust between workers and clients by ensuring transparency regarding worker personal details.
- [0036]** FIG. 2 illustrates a flowchart for a job posting and discovery method 200 according to various embodiments. The method 200 depicts parallel workflows for clients posting jobs and workers discovering and applying for jobs.
- [0037]** The method 200 begins with a step 202, where a job process is initiated. The method 200 then proceeds to a step 204, which involves a determination of a user role, specifically whether the user is a client or a worker. Based on this determination, the method 200 branches into one of two distinct workflow paths.
- [0038]** When the user is identified as a client at step 204, the method 200 proceeds along a client branch. A step 206 involves entering job details and specifications. During step 206, the client specifies job posting parameters

including job title, description, required skills, wage, work location, time duration, and gender preference. The client sets application deadlines and defines a maximum number of workers required for the job posting. The job posting parameters enable targeted and efficient hiring by providing workers with detailed information about job requirements.

[0039] The method 200 then moves to a step 208, where a job ID is assigned and job information is stored in a database. During step 208, each job posting receives a unique identifier for tracking and future reference. The database stores the job posting in a structured format that supports efficient queries and geolocation filtering. The database is a NoSQL database that supports fast queries and location-based filtering operations.

[0040] Following step 208, a step 210 triggers notifications to matching workers. During step 210, the system sends notifications to workers whose profiles match the job posting parameters. The notifications alert workers to new job postings matching their profile, enabling workers to discover relevant opportunities in a timely manner. The system incorporates an intelligent job-matching mechanism that suggests workers for a given job based on relevant parameters including skills, location, and availability. The client branch concludes at a step 212, where the job posting is marked as complete.

[0041] When the user is identified as a worker at step 204, the method 200 proceeds along a worker branch. A step 214 involves accessing a job listings dashboard. The job listings dashboard provides workers with a structured interface for browsing available job opportunities.

[0042] The method 200 then moves to a step 216, where a location-based filter is applied. During step 216, a dynamic location-based filtering system displays job opportunities within the worker's city by default. The city-default filtering ensures that workers focus on nearby, relevant jobs without being overwhelmed with job listings from across the country. The location-based filtering addresses the challenge of workers receiving irrelevant or inaccessible job listings by presenting geographically appropriate opportunities.

[0043] The method 200 proceeds to a step 218, which is a determination as to whether additional filters should be applied. If additional filters are to be applied, the method 200 proceeds to a step 220, where filtering by wage, job type, and region is performed. During step 220, workers browse jobs based on location proximity, wage preferences, work category, and job type using a filtering system. The filtering system provides options to apply advanced filters to explore work opportunities across other states or regions beyond the default city-based display. The multi-parameter filtering enables workers to refine job searches based on specific criteria including pay range, work type, and geographic scope.

10 **[0044]** If no additional filters are required at step 218, or after step 220 is completed, the method 200 proceeds to a step 222, where filtered job listings are displayed. During step 222, jobs are displayed in a structured format that allows workers to browse and evaluate opportunities efficiently. The filtered job listings present job information including wage, location, required skills, duration, and gender preferences.

15 **[0045]** The method 200 then moves to a step 224, which is a determination as to whether the worker wishes to apply for a job. If the worker chooses to apply, the method 200 proceeds to a step 226, where an application is submitted and saved to applied jobs. During step 226, workers apply for jobs with a single click. A confirmation screen ensures that workers understand job details before submission. The one-click application process eliminates the need for in-person job searches by providing a streamlined digital application mechanism. Workers withdraw applications before selection by the client, providing flexibility in managing job applications. Once applied, jobs are saved in the worker's applied list for tracking purposes.

20 **[0046]** If the worker chooses not to apply at step 224, the method 200 returns to step 222 to continue displaying filtered job listings, enabling the worker to evaluate additional opportunities.

25 **[0047]** The method 200 provides parallel workflows that connect clients posting jobs with workers discovering and applying for jobs. The location-based filtering serves as a default mechanism that presents geographically relevant

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opportunities, while the optional multi-parameter filtering enables workers to refine searches based on wage, job type, and region parameters. The notification system and intelligent job-matching mechanism facilitate efficient connections between job postings and qualified workers.

5 **[0048]** FIG. 3 illustrates a flowchart for a job assignment and identity verification method 300 according to various embodiments. The method 300 provides a dual-layer authentication mechanism that combines profile review with on-site identity verification using Aadhaar-based photo cross-verification.

10 **[0049]** The method 300 begins with a step 302, where a client receives job applications from workers who have applied for a posted job listing. During step 302, the client accesses applications submitted by workers through the platform. The applications include worker profile information, skills, experience, and past job history.

15 **[0050]** The method 300 then proceeds to a step 304, where the client reviews worker profiles and shortlists suitable candidates based on job requirements. During step 304, the client reviews applicants, views profiles and past reviews, and shortlists candidates based on job requirements. The client evaluates worker profiles that include details such as skills, experience, past job history, availability, and client ratings. The review process enables the client to assess worker credibility
20 based on ratings and reviews from previous engagements. The shortlisting functionality allows the client to identify candidates whose qualifications align with the posted job specifications.

25 **[0051]** Following step 304, the method 300 moves to a step 306, where the client selects a worker and sends a notification to inform the selected worker of the selection. During step 306, selected workers receive notifications confirming their selection for a job. The notification system provides real-time communication between the client and the selected worker regarding job assignment status.

30 **[0052]** The method 300 then reaches a step 308, which involves a determination as to whether the worker accepts the job. If the worker accepts the job, the method 300 proceeds to a step 310, where the worker arrives at a job location. If the worker does not accept the job, the method 300 returns to step 304

to review worker profiles and shortlist alternative candidates. The acceptance decision enables workers to evaluate job details before committing to an assignment.

5 **[0053]** Upon the worker arriving at the job location at step 310, the method 300 continues to a step 312, where the client captures a photo of the worker. During step 312, the client uses a device to photograph the worker at the job site. The photo capture forms the first component of the on-site identity verification process.

10 **[0054]** The method 300 then advances to a step 314, where a system cross-verifies the captured photo with an Aadhaar image previously uploaded by the worker during registration. During step 314, a facial recognition model trained to compare live images with Aadhaar photographs performs real-time identity verification at the job site. The cross-verification compares facial features from the captured photo against the Aadhaar card photograph stored in the database during the worker registration process described in the method 100. The dual-layered authentication confirms the identity of the worker by validating that the person
15 present at the job location matches the registered worker profile.

20 **[0055]** The method 300 then reaches a step 316, which involves a determination as to whether identity is verified. If the identity is successfully verified, the method 300 proceeds to a step 318, where the system confirms the assignment and updates a job status accordingly. During step 318, the job assignment is confirmed and the job status transitions to an active state. The status update reflects in tracking systems accessible to both the client and the worker.

25 **[0056]** If the identity verification fails at step 316, the method 300 moves to a step 320, where a worker profile is flagged indicating verification failure. During step 320, the system records the verification failure and associates a flag with the worker profile. The flagging mechanism provides accountability and enables the platform to track verification discrepancies for review.

30 **[0057]** Following successful identity verification and assignment confirmation at step 318, the method 300 concludes with a step 322, where the worker begins the job. Assigned workers view accepted jobs in a My Jobs section for tracking purposes. The My Jobs section provides workers with a dedicated dashboard to

monitor assigned jobs, track job progress, and manage active engagements. Clients track job progress and communicate through an in-app messaging feature, enabling coordination between the client and the worker during job execution.

[0058] The method 300 provides a dual-layer authentication mechanism that enhances trust between clients and workers. The first layer involves profile review during the shortlisting process at step 304, where clients evaluate worker credentials, past reviews, and ratings. The second layer involves on-site identity verification at steps 312 through 316, where the captured photo is cross-verified against the Aadhaar photograph using facial recognition. The combination of profile-based evaluation and biometric verification reduces the potential for impersonation and establishes confidence in worker-client relationships within the platform.

[0059] FIG. 4 illustrates a flowchart for a job completion and review method 400 according to various embodiments. The method 400 provides a workflow for task completion verification, payment processing, and bidirectional rating between workers and clients.

[0060] The method 400 begins with a step 402, where a worker performs tasks and updates status. During step 402, the worker executes assigned job tasks and provides status updates through the platform. The status updates enable clients to track job progress in real-time. Workers update task status by marking jobs as in progress or completed, providing visibility into job execution stages.

[0061] The method 400 then proceeds to a step 404, where the worker submits completion proof. During step 404, upon completing a job, the worker submits proof such as a photo or task completion update. The completion proof provides documentation that the assigned tasks have been performed. The photo submission or task completion update serves as evidence of job execution that the client reviews to verify work completion.

[0062] Following step 404, the method 400 moves to a step 406, where the client provides a verification code. During step 406, the client generates and communicates a verification code to the worker. The verification code serves as a

confirmation mechanism that validates successful job completion from the client's perspective.

[0063] The method 400 then reaches a step 408, which involves a determination as to whether the verification code is correct. During step 408, the worker enters the client-provided verification code into the platform. The system validates the entered code against the code provided by the client. If the verification code is not correct, the method 400 returns to step 406 to receive the verification code again from the client. When the verification code is incorrect, the job is flagged as Pending Confirmation, indicating that the completion verification process requires additional attention. If the verification code is correct, the method 400 proceeds to a step 410, where the job is marked as completed. During step 410, the system updates the job status to reflect successful completion and records the completion timestamp.

[0064] The method 400 then moves to a step 412, which involves a determination of a payment method, specifically whether the payment method is cash or digital. The platform supports dual payment methods that provide flexibility for both workers and clients in settling job compensation.

[0065] If the payment method is cash at step 412, the method 400 proceeds to a step 414, where a cash payment is recorded. During step 414, the system records the cash transaction details including the payment amount and transaction date. The cash payment recording maintains a record of compensation provided to the worker for tracking and accounting purposes. The method 400 then moves to a step 418, where earnings are updated and reviews are initiated. During step 418, the worker's earnings dashboard is updated to reflect the cash payment received. The earnings dashboard enables workers to track earnings over different time periods, providing visibility into compensation history across multiple jobs.

[0066] If the payment method is digital at step 412, the method 400 proceeds to a step 416, where payment is processed via an in-app payment gateway. During step 416, clients pay workers via the in-app payment gateway with secure digital transactions ensuring transparency. The in-app payment gateway processes the digital transaction and transfers funds to the worker. The platform supports UPI-

based wallet integrations for secure and traceable payments, providing a digital payment infrastructure that maintains transaction records and enables verification of payment completion. Workers receive immediate payment notifications upon successful transactions, confirming that the digital payment has been processed and funds have been transferred. The method 400 then moves to a step 420, where earnings are updated and reviews are initiated. During step 420, the worker's earnings dashboard is updated to reflect the digital payment received. The earnings dashboard provides workers with the capability to track earnings over different time periods, enabling workers to monitor compensation across daily, weekly, monthly, or custom date ranges.

[0067] Following step 418, the method 400 proceeds to a step 422, where bidirectional rating is complete. Following step 420, the method 400 proceeds to a step 424, where bidirectional rating is complete. Steps 422 and 424 represent the conclusion of the bidirectional rating process for the cash and digital payment paths respectively.

[0068] During the review initiation at steps 418 and 420, the platform enables both clients and workers to evaluate each other following job completion. Clients rate workers based on punctuality, work quality, and professionalism. The client ratings aid future clients in selection by providing performance indicators that reflect worker reliability and job execution quality. The rating criteria enable clients to assess specific aspects of worker performance, contributing to worker credibility within the platform.

[0069] Workers rate clients based on behavior, fairness in payment, and communication. The worker ratings provide feedback regarding client conduct during job engagements. A flagging system enables workers to report issues such as payment disputes or inappropriate behavior. The flagging system provides a mechanism for workers to document concerns and escalate issues that require platform review. The bidirectional rating system fosters transparency and accountability by enabling both parties to provide feedback that informs future engagements within the platform.

[0070] The method 400 provides a structured workflow that addresses job completion verification through proof submission and verification code confirmation, supports dual payment methods through cash recording and digital payment gateway processing, and implements a bidirectional rating system that enables both workers and clients to evaluate each other. The earnings dashboard provides workers with comprehensive tracking of compensation across different time periods, while the flagging system enables reporting of issues that arise during job engagements.

[0071] The platform is built using a technology stack that includes Node.js for backend development and MongoDB for database operations. The technology stack provides a scalable and efficient infrastructure for managing worker and client profiles, job postings, and dynamic search functionalities.

[0072] A frontend of the platform is built using React.js with a component-based architecture. The component-based architecture enables efficient development and code reusability by breaking down a user interface into smaller, reusable components. The components are reused across different parts of the application, reducing development time and ensuring consistency across the platform.

[0073] A backend of the platform uses Express.js as a web application framework for Node.js. Express.js simplifies route handling and middleware management by providing an efficient mechanism for defining routes that handle different types of HTTP requests including GET, POST, PUT, and DELETE operations. Express.js enables middleware functions to handle tasks including logging, authentication, error handling, and data validation.

[0074] MongoDB serves as a NoSQL database that stores data in a flexible, JSON-like format. MongoDB is used with Mongoose as an Object Document Mapper (ODM) for data validation and schema design. Mongoose simplifies interactions with the database by providing schema definitions, data validation rules, and query building capabilities. The Mongoose ODM integration enables efficient storage and retrieval of unstructured or semi-structured data including user profiles, job listings, worker applications, and review records.

[0075] A centralized database securely stores all user details, job listings, worker applications, task statuses, and reviews. Real-time data updates ensure all modules remain synchronized, facilitating a seamless user experience across the platform. The centralized database maintains data consistency by synchronizing changes across authentication, job management, and worker management modules.

[0076] The platform uses WebSockets to facilitate real-time communication between a server and a client. WebSocket connections enable real-time features including notifications, job alerts, and status updates to be delivered without polling delays. A library such as Socket.io provides WebSocket functionality that supports bidirectional communication channels between the server and connected clients.

[0077] Caching mechanisms including Redis are implemented to reduce database load and speed up data retrieval. Redis caching stores frequently accessed data in memory, enabling rapid retrieval without repeated database queries. The caching mechanism improves response times for operations including job searches and profile lookups by serving cached data when available.

[0078] The platform is deployed on Vercel, which offers streamlined deployment with automatic scaling, serverless functions for APIs, and seamless integration with Git repositories. Vercel provides a serverless deployment infrastructure that handles frontend hosting and backend logic through serverless function execution. The Vercel deployment enables the platform to deploy the frontend and handle backend operations including user requests and data management without dedicated server infrastructure.

[0079] The platform uses Cloudinary for cloud-based image storage and management of uploaded photos and documents. Cloudinary provides storage and retrieval services for profile pictures, Aadhaar card images, and job completion proof photographs. The platform supports multiple image formats including PNG with specific handling options for different file types during upload. Image upload handling includes format detection, buffer validation, and format-specific configuration options that ensure proper processing of various image types.

[0080] User data is encrypted in transit using TLS 1.2 or higher. Sensitive personal and financial data is stored with AES-256 encryption. The TLS encryption

protects data during transmission between clients and servers, while AES-256 encryption secures stored data against unauthorized access.

[0081] The platform employs protection against SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). Security measures include
5 input validation, output encoding, and token-based request verification that prevent common web application vulnerabilities. Regular security audits and penetration testing identify and address potential vulnerabilities within the platform.

[0082] Load balancing is implemented to distribute incoming traffic evenly across servers, preventing any single server from being overwhelmed. The load
10 balancing mechanism routes incoming requests to available server instances based on current load conditions, ensuring consistent response times during traffic spikes.

[0083] Auto-scaling features are utilized to dynamically adjust a number of servers based on current demand. The auto-scaling mechanism monitors resource utilization metrics and provisions additional server instances when demand
15 increases, while reducing server count during periods of lower activity.

[0084] The system implements graceful degradation mechanisms including rate limiting and fallback responses to ensure the platform remains responsive under extreme stress. Rate limiting restricts the number of requests from individual clients within specified time windows, preventing resource exhaustion from
20 excessive request volumes. Fallback responses provide alternative functionality when primary services experience degradation, maintaining platform availability during high-load conditions.

[0085] The platform is configured to load a homepage within two seconds under normal network conditions. Job search and application submissions are
25 configured to complete within three seconds. The response time requirements ensure responsive user interactions across browsing, searching, and application submission operations.

[0086] The system is designed to support at least 10,000 concurrent users without performance degradation. The concurrent user capacity is achieved through
30 the combination of load balancing, auto-scaling, caching mechanisms, and

optimized database queries that distribute processing load across available resources.

[0087] The platform is configured to maintain 99.9% uptime, ensuring that functionalities including job applications and payment processing remain operational during peak usage periods. Redundancy mechanisms including database replication and backup and recovery strategies ensure platform availability during server failures. Automated failover and real-time error handling maintain platform stability by redirecting traffic and recovering from failure states.

[0088] The platform includes a suspension history feature that tracks and displays worker suspension records. Suspension records include date, duration, reason, and additional details associated with each suspension event. The suspension history provides a chronological record of disciplinary actions that enables platform administrators and users to review past suspension events.

[0089] A notes management system allows users to create, edit, delete, and filter notes with priority levels and date ranges. The notes management system supports priority classification including high, medium, and low priority designations. Users filter notes based on creation date ranges and priority levels to locate specific records. The notes functionality enables users to maintain records and reminders associated with job activities and platform interactions.

[0090] Working Example 1 demonstrates a complete worker registration flow with Aadhaar validation and OTP verification. A worker initiates registration through the platform and selects worker registration as the user type. The worker enters personal details including name "Rajesh Kumar," phone number "9876543210," gender "Male," location "Mumbai, Maharashtra, India," and work category "Plumber." The worker specifies skills including "pipe repair," "leak detection," and "fixture installation" to improve job matching accuracy. The worker sets availability status to "actively looking for jobs."

[0091] The worker uploads a profile picture in PNG format and an Aadhaar card image for identity verification. The platform processes the Aadhaar card image using a machine learning integration for document parsing. The document parsing employs image processing and Optical Character Recognition (OCR) to

automatically extract and populate user data from the uploaded Aadhaar card. The OCR module processes the Aadhaar card image and extracts data fields including name, date of birth, address, and Aadhaar number. The extracted data is validated against the manually entered personal details to verify consistency. The machine
5 learning-based document parsing reduces onboarding friction by automatically populating form fields and improves data accuracy by extracting information directly from the official identity document.

[0092] The platform sends an OTP to the phone number "9876543210" provided by the worker. The worker receives the OTP "847291" via SMS and enters
10 the OTP into the verification field. The system validates the entered OTP against the generated OTP and confirms successful verification. The platform encrypts the worker credentials using bcrypt password hashing and stores the encrypted credentials in the MongoDB database. The system assigns role-based access control for worker functionality and generates a JWT token with a payload containing the
15 worker user ID and role designation. The worker registration completes with the worker profile stored in the database and the JWT token returned for subsequent authenticated requests.

[0093] Working Example 2 demonstrates a job posting and discovery workflow with location-based filtering. A client initiates job posting through the
20 platform and enters job details including job title "Bathroom Pipe Repair," description "Repair and replace leaking pipes in bathroom and kitchen," required skills "plumbing, pipe repair," wage range "₹500-₹1000 per day," work location "Mumbai, Maharashtra, India, 400001," duration "3 days," start date "January 05, 2025," reporting time "09:00 AM," gender preference "any," and number of
25 vacancies "2." The platform assigns a unique job ID "JOB-2025-00847" and stores the job information in the MongoDB database with geolocation coordinates derived from the specified location.

[0094] The platform triggers notifications to matching workers based on job parameters. A machine learning-based job recommendation engine analyzes
30 historical applications, success rates, and user preferences to identify workers whose profiles match the job posting. The recommendation engine evaluates

worker skill sets, past job completion rates, client ratings, and geographic proximity to generate a ranked list of matching workers. Workers matching the job criteria receive push notifications alerting them to the new job posting.

5 **[0095]** A worker "Rajesh Kumar" accesses the job listings dashboard and views job opportunities. The platform applies a location-based filter that displays job opportunities within the worker's city "Mumbai" by default. The worker views the "Bathroom Pipe Repair" job listing among the filtered results. The worker applies additional filters to refine the search, selecting wage range "₹400-₹1200 per day" and job type "Plumber." The filtered job listings display jobs matching the
10 specified criteria within the Mumbai region.

[0096] The worker selects the "Bathroom Pipe Repair" job and reviews job details including wage, location, duration, and required skills. The worker clicks the apply button and views a confirmation screen displaying job details. The worker confirms the application, and the platform submits the application and saves the job
15 to the worker's applied jobs list. The client receives a notification indicating that worker "Rajesh Kumar" has applied for the "Bathroom Pipe Repair" job posting.

[0097] Working Example 3 demonstrates the identity verification and job completion process with bidirectional rating. The client reviews applications for the "Bathroom Pipe Repair" job and views the profile of worker "Rajesh Kumar." The
20 client evaluates the worker profile including skills "pipe repair, leak detection, fixture installation," experience "5 years," past job history "47 completed jobs," availability "available," and client ratings "4.7 out of 5 stars based on 42 reviews." The client shortlists "Rajesh Kumar" based on the profile evaluation and selects the worker for the job assignment. The platform sends a notification to "Rajesh Kumar"
25 confirming selection for the "Bathroom Pipe Repair" job.

[0098] The worker "Rajesh Kumar" accepts the job and arrives at the job location on January 05, 2025 at 09:00 AM. The client captures a photo of the worker using a mobile device camera. The platform cross-verifies the captured photo with the Aadhaar image uploaded during worker registration. A facial recognition model
30 compares facial features from the captured photo against the Aadhaar card

photograph stored in the database. The system confirms identity verification with a confidence score of 94.7% and updates the job status to "in progress."

[0099] The worker performs the assigned tasks over the 3-day duration and updates task status to "completed" upon finishing the work. The worker submits completion proof including a photo of the repaired pipes. The client reviews the completion proof and provides verification code "7829" to the worker. The worker enters the verification code into the platform, and the system validates the code and marks the job as completed.

[0100] The client selects digital payment and processes payment of ₹800 via the in-app payment gateway using UPI. The worker receives an immediate payment notification confirming the ₹800 transfer. The worker's earnings dashboard updates to reflect the payment received.

[0101] The platform initiates the bidirectional rating process. The client rates the worker "Rajesh Kumar" with 5 stars for punctuality, 5 stars for work quality, and 4 stars for professionalism, with a comment "Excellent work, completed ahead of schedule." The worker rates the client with 5 stars for behavior, 5 stars for fairness in payment, and 5 stars for communication, with a comment "Clear instructions and prompt payment." The bidirectional ratings are stored in the database and update the respective user profiles.

[0102] Experimental validation was conducted using Apache JMeter to assess platform performance under load conditions. Stress and load testing evaluated how the platform handles increasing traffic and concurrent API requests on endpoints including user registration, job posting, job search, and login. Test scenarios simulated real-world usage patterns including surge conditions during peak usage periods.

Endpoint	Concurrent Users	Loops	Delay (seconds)	Average Response Time (ms)	Throughput (requests/second)	Standard Deviation (ms)	Error Rate (%)
GET /api/jobs	1000	4	2	847	312.4	156	0.12

POST /api/auth/login	1000	4	2	923	287.6	189	0.08
POST /api/jobs/apply	1000	4	2	1124	245.3	234	0.15
GET /api/workers/profile	1000	4	2	756	334.8	142	0.05
POST /api/jobs/create	1000	4	2	1089	258.7	218	0.11

[0103] The load testing results demonstrate that the platform maintains response times within acceptable thresholds under 1000 concurrent users at 4 loops with 2-second delay. The GET /api/jobs endpoint achieved throughput of 312.4 requests per second with average response time of 847 milliseconds and standard deviation of 156 milliseconds. The POST /api/auth/login endpoint achieved throughput of 287.6 requests per second with average response time of 923 milliseconds. Error rates remained below 0.2% across all tested endpoints, indicating stable operation under load conditions.

[0104] Query optimization techniques were applied to endpoints exhibiting latency beyond acceptable thresholds. Database indexes were refined for job search queries involving location-based filtering and multi-parameter filtering operations. API logic was restructured to reduce database round trips and improve query efficiency. Following optimization, average response times decreased by approximately 35% for complex filtered job searches.

[0105] The platform incorporates multilingual voice assistant support to aid workers with limited literacy. The voice assistant provides spoken instructions and guidance in regional languages including Hindi, Tamil, Telugu, Marathi, and Bengali. Workers interact with the platform using voice commands to browse job listings, apply for jobs, and check application status. The voice assistant converts

spoken input to text using speech recognition and provides audio responses using text-to-speech synthesis. The multilingual support addresses language barriers that limit platform adoption among workers with limited reading proficiency.

5 [0106] Administrative dashboards are provided for NGOs and government bodies to track employment trends and develop targeted welfare schemes. The administrative dashboards display aggregated employment data including job posting volumes, application rates, completion rates, and wage distributions across geographic regions. The dashboards provide visualization of employment trends over time periods including daily, weekly, monthly, and annual intervals. Filtering capabilities enable administrators to analyze data by region, job category, worker demographics, and wage ranges. The administrative dashboards support export of employment data in formats including CSV and JSON for integration with external analysis tools and welfare program planning systems.

10 [0107] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This 27th day of February 2026

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(Kuldeep Singh)

Authorized Agent for the Applicant,

Indian Patent Agent Regn No. IN/PA-4358

WE CLAIM:

1. A system for connecting daily wage workers with job opportunities, the system comprising:

5 a server configured to execute instructions stored in a memory, the server comprising:

a user registration module configured to receive registration data from a plurality of users, wherein each user is classified as either a worker or a client, wherein for workers the registration data comprises an identity document image;

10 a database configured to store the identity document image associated with each worker;

a job assignment module configured to enable assignment of workers to jobs; and

an identity verification module configured to:

receive a captured image of a worker at a job location from a client device;

15 cross-verify the captured image with the identity document image stored during worker registration; and

generate a verification confirmation upon successful cross-verification.

2. The system of claim 1, wherein the server further comprises an authentication module configured to perform OTP-based phone number
20 verification for each user, encrypt user credentials, and store encrypted credentials in the database.

3. The system of claim 1 or 2, wherein the server further comprises a job discovery module configured to display job listings to workers, wherein the job discovery module is configured to apply a location-based filter to display jobs
25 within a worker's city by default.

4. The system of any of claims 1 to 3, wherein the identity verification module is further configured to extract identity information from the identity document image using optical character recognition.

5. The system of any of claims 1 to 4, wherein the identity verification
30 module is further configured to flag a worker profile upon cross-verification failure.

6. The system of any of claims 1 to 5, wherein the server further comprises:

a job completion module configured to receive task completion proof from a worker, receive a verification code from a client, validate the verification code, and mark a job as completed upon successful validation; and

5 a review module configured to receive ratings and reviews from clients regarding worker performance and receive ratings and reviews from workers regarding client behaviour.

7. A computer-implemented method (100, 200, 300) for connecting daily wage workers with job opportunities using dual-layer identity verification, the method comprising:

10 receiving (106, 108), by a server, registration data from a user classified as a worker, wherein the registration data comprises an identity document image;

storing (114) the identity document image in a database;

assigning (306) the worker to a job;

15 receiving (312), from a client device, a captured image of the worker at a job location;

cross-verifying (314) the captured image with the identity document image stored during worker registration; and

generating (318) a verification confirmation upon successful cross-verification.

20 8. The method of claim 7, further comprising:

performing (110, 112) OTP-based phone number verification for the user;

and

encrypting (114) user credentials and storing encrypted credentials in the database.

25 9. The method of claim 7 or 8, further comprising:

displaying (222) job listings to the worker with a location-based filter (216) applied to display jobs within a worker's city by default; and

receiving (220) filter parameters from the worker and displaying filtered job listings based on received filter parameters.

10. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to perform the method of any of claims 7 to 9.

Dated This 27th day of February 2026

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A handwritten signature in dark ink, appearing to read 'Kuldeep', is written over a horizontal line.

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ABSTRACT

LOCATION-FILTERED SERVICE PLATFORM WITH BIDIRECTIONAL RATING AND IMAGE VERIFICATION

The present disclosure provides a system for connecting daily wage workers with
5 job opportunities. The system includes a server (100) configured to execute
instructions stored in a memory. The server includes a user registration module
(102, 104, 106, 108) configured to receive registration data from users classified as
either workers or clients, wherein worker registration data includes an identity
document image. The server includes a database (114) configured to store identity
10 document images associated with each worker. The server includes a job
assignment module (306) configured to enable assignment of workers to jobs. The
server includes an identity verification module (312, 314, 316, 318) configured to
receive a captured image of a worker at a job location from a client device, cross-
verify it with the stored identity document image, and generate a verification
15 confirmation upon successful cross-verification.

(FIG. 1)

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